Won Jun Lee

Ph.D. Candidate in Electrical and Computer Engineering, Purdue University

Profile

Ph.D. candidate researcher in network scheduling, stochastic optimization, and scalable algorithm design for resource allocation, low-latency communication, and **information freshness** in networked systems. Develops provably optimal scheduling policies for **multi-channel communication** and **multi-sensor estimation**, and turns structural properties of the optimal policy into algorithms that scale to large systems.

Contact Details

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Website

Professional Service

Reviewer, IEEE International Symposium on Information Theory (ISIT), *2026*

Awards & Honors

Ross Fellowship, Purdue University, *2021–2025*
Student Travel Grant, IEEE International Symposium on Information Theory (ISIT), *2024*
Dean's List, Yonsei University, *2015–2016, 2019–2020*

Skills

Modeling & Optimization: *Convex Optimization, Stochastic Processes, Probability Theory, Information Theory, Control Theory, Graph Theory*
Programming Languages: *MATLAB, C, Python, Java/Android*

Education

Purdue University — *West Lafayette, IN, U.S.*
Ph.D. Candidate, Electrical and Computer Engineering — *Aug 2021 – Exp. May 2027*
Advisor: Prof. Chih-Chun Wang
GPA: 3.81/4.0

Yonsei University — *Seoul, South Korea*
B.S., Electrical and Electronics Engineering — *Mar 2015 – Feb 2021*
GPA: 3.84/4.0

Research Experience

Prof. Chih-Chun Wang's Lab
West Lafayette, IN, U.S.
Research Assistant, Purdue University — *Aug 2021 – Present*

Latency-and-Cost-Aware Channel Scheduling → [C2]

- Developed a **scheduling and resource-allocation framework** for heterogeneous multi-channel systems, jointly optimizing latency, transmission cost, data quality, and **information freshness**.
- Designed a **structured channel-selection algorithm** that systematically computes the policy thresholds and enables scalable scheduling decisions in systems with diverse channel attributes.
- **Reduced the computational complexity** of finding the scheduling policy from $O(N^3)$ to $O(N)$, improving the practicality of the approach for larger-scale systems.

Cost-Aware Sensor Scheduling in the Correlated Noise Environment

- Developed a **cost-aware scheduling framework** for multi-sensor systems under **correlated noise**, targeting efficient estimation under communication and sensing constraints.
- Designed an estimation method that **alleviates the computational burden** associated with **Kalman filtering** in correlated-noise settings, improving tractability for high-dimensional systems.
- Proposed an optimization approach for balancing estimation error and communication cost in **resource-constrained sensing systems**.

Digital Communications Lab
Seoul, South Korea
Undergraduate Research Intern, Yonsei University — *Jan – Jun 2020*

5G Network Performance Benchmark in Heterogeneous Environment

- Developed an Android-based network analysis application using **UDP socket programming** to measure end-to-end latency/throughput in heterogeneous network communication environments.
- Implemented **UDP hole punching** to support connectivity benchmarking in real-world 5G settings.

High Dimensional Signal Processing Lab
Seoul, South Korea
Undergraduate Research Intern, Yonsei University — *Jul – Aug 2019*

DNN Classification Logic Analysis → [C1]

- Assisted in analyzing decision boundaries of deep neural network classifiers.

Military Experience

KATUSA (Korean Augmentation To the United States Army) — *Seoul, South Korea*
Sergeant — *Mar 2017 – Dec 2018*

- Served in the Information Management Officer duty position (MOS 25B, IT Specialist) at the USFK Public Affairs Office, supporting mission-critical IT and communications operations.
- Led 6 soldiers as a **squad leader** while managing approximately 100 network-connected workstations and coordinating technical operations between ROK and U.S. military teams during **international joint military exercises** (Ulchi-Freedom Guardian 2017 / Foal Eagle 2018).

Publications

- **[C2] W.J. Lee** and C.-C. Wang, "AoI-optimal scheduling for arbitrary K-channel update-through-queue systems," *in Proc. IEEE Int. Symp. on Information Theory (ISIT)*, 2024, pp. 957–962.
- **[C1]** J. Kim, S. Woo, **W.J. Lee**, D. Kim, and C. Lee, "Analyzing Decision Polygons of DNN-based Classification Methods," Proc. 17th Int. Conf. Informatics in Control, Automation and Robotics (ICINCO), 2020, pp. 112–119